



## ENT-164 · Intro to Making

BEGINNER'S GUIDE · MACOS & WINDOWS · NO EXPERIENCE NEEDED

# Making with AI

## OpenCode + DeepSeek V4.1 Flash

Download the OpenCode app, sign in to the class workspace, and start asking questions. No terminal, no coding — just a normal app.

About 20 minutes

For macOS & Windows

Provider: OpenCode Console

Model: DeepSeek V4.1 Flash

Ask anything, / for commands, @ for context...

+  DeepSeek V4.1 Flash ▾ Default ▾



The OpenCode message box, with DeepSeek V4.1 Flash already selected. This is where you type your questions.

**OpenCode** is an AI helper that runs as an app on your computer. You type to it in plain English, and it can explain code, help with class projects, and walk you through technical tasks. **DeepSeek V4.1 Flash** is the AI "brain" that writes the answers, and **OpenCode Console** is the service that connects the app to it.

Think of it this way: OpenCode is your phone, OpenCode Console is your phone company, and DeepSeek V4.1 Flash is the person you're calling. This guide sets all three up. You won't need to use a terminal or write any code.

#### Why DeepSeek V4.1 Flash for this class?

- **Built for this work** — it's designed for reasoning and agentic coding with reliable tool use, so it can genuinely help with your class projects.
- **Open weights** — the model itself is openly licensed and inspectable, which fits a making class better than a closed black box.
- **Big context, tiny cost** — it can read entire files (up to 1M tokens) for a fraction of a cent per question, so our shared class budget lasts the term.
- **One connection** — because Console gives us access to it, you set up one account and one sign-in, and you can explore other models later without starting over.

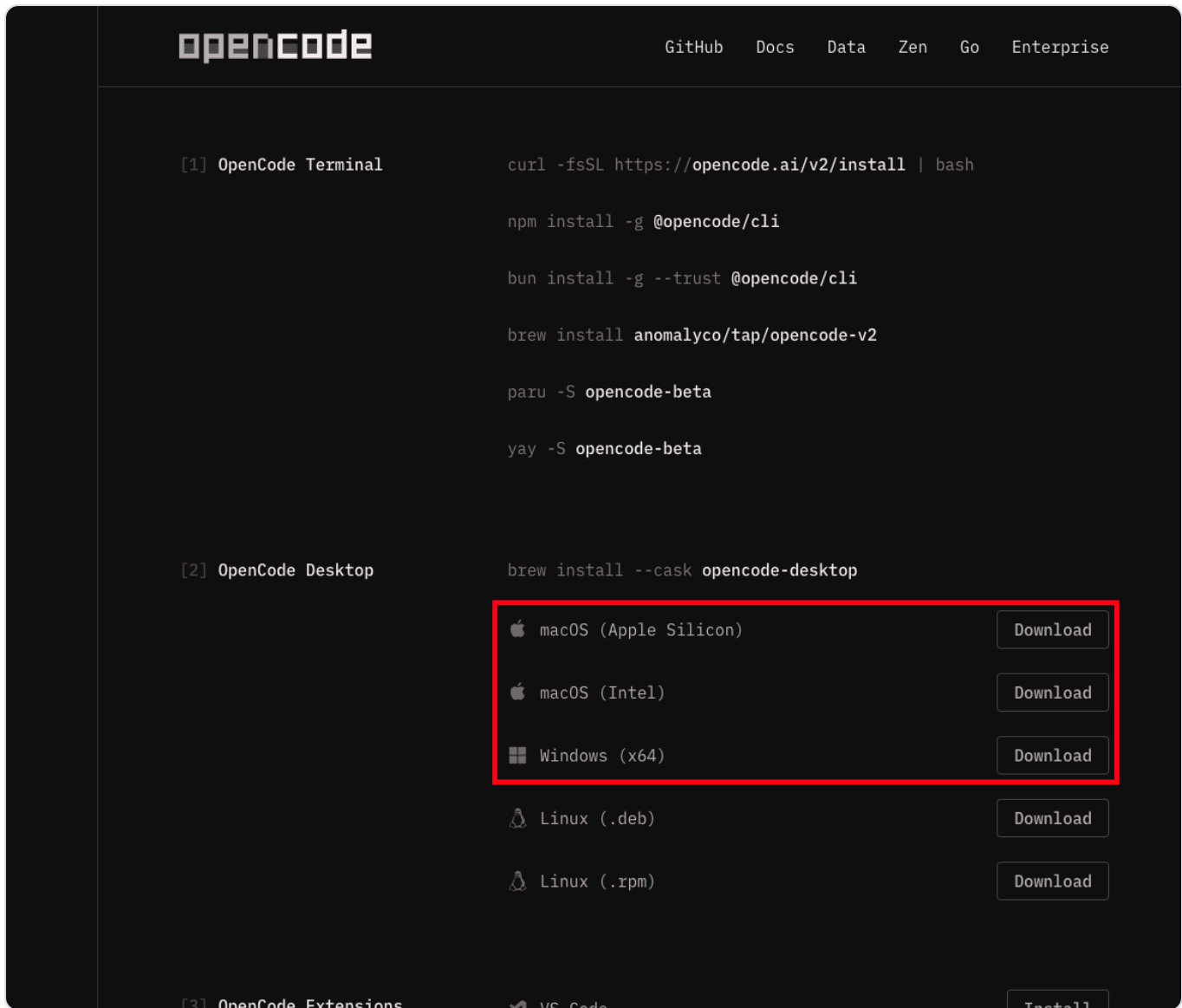
#### What you'll need

- ✓ A laptop running macOS 13 or newer, or Windows 10 or 11
- ✓ An internet connection
- ✓ Your Tufts email — the class invitation goes to it, and you sign in with it in Step 3
- ✓ Nothing to pay — the class budget covers AI usage for your coursework

**New word alert.** **OpenCode Console** is the OpenCode team's service that gives one account access to a set of AI models they have tested and chosen (you may also see it called **OpenCode Zen** in older class notes — it's the same service). The app calls it **OpenCode Console** everywhere, so that's the name this guide uses.

## 1 Download the OpenCode app

1. Open your browser and go to **[opencode.ai/download](https://opencode.ai/download)**.
2. Under **OpenCode Desktop**, click the download for your computer: **macOS** (Apple Silicon or Intel) or **Windows (x64)**.
3. The download starts — you'll find the file in your **Downloads** folder.



**Figure 1.** The download page. Under **OpenCode Desktop**, pick the installer for your computer — the page lists both macOS and Windows.

## 2 Install the app

### On a Mac:

1. Open the downloaded file (click it in your **Downloads** folder, or in the notification Safari shows).
2. A small window opens showing the **OpenCode** icon and your **Applications** folder.
3. **Drag the OpenCode icon onto the Applications folder**, exactly like the arrow shows.
4. Close that window, then open **OpenCode** from your Applications folder (or press  + , type **OpenCode**, and press ).

### On Windows:

1. Open your **Downloads** folder and double-click the OpenCode installer (the file ends in **.exe**).

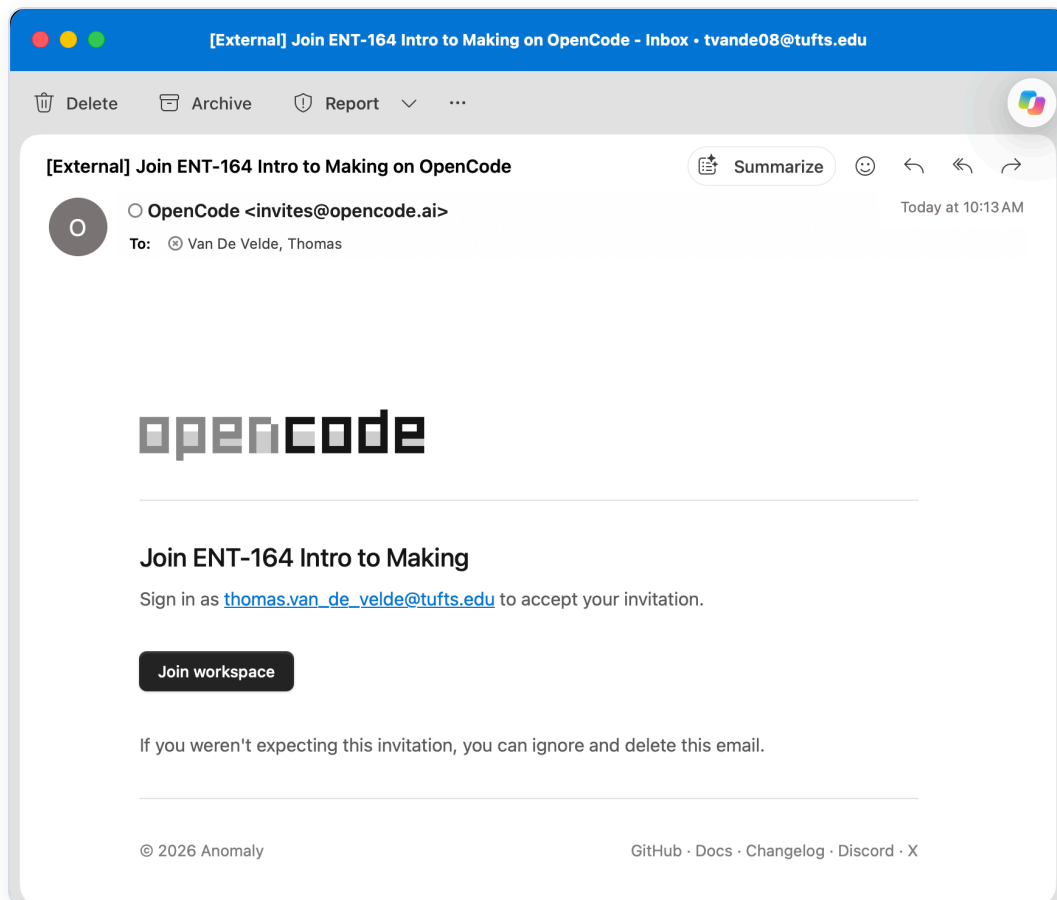
2. Follow the prompts. If Windows shows "**Windows protected your PC**", click **More info**, then **Run anyway**.
3. When it finishes, open OpenCode from the **Start menu**: press the **Windows** key, type **OpenCode**, and press **Enter**.

**Your computer warns you about the app?** That's the standard message for apps downloaded from the internet. On a Mac, right-click (or Control-click) **OpenCode** in Applications, choose **Open**, and confirm. On Windows, click **More info** → **Run anyway**. The app is signed and comes directly from `opencode.ai`.

### 3 Accept your class invitation

Your instructor adds you to the class workspace on OpenCode Console using your **Tufts email**. The invitation arrives by email, and you accept it once.

1. Check your Tufts inbox for an invitation to the class workspace and click the link in it.
2. Sign in with your **Tufts email** (for example `yourname@tufts.edu`). Using the **Google** button skips creating a separate password.
3. You land in the class workspace — it's named **ENT-164 Intro to Making**.



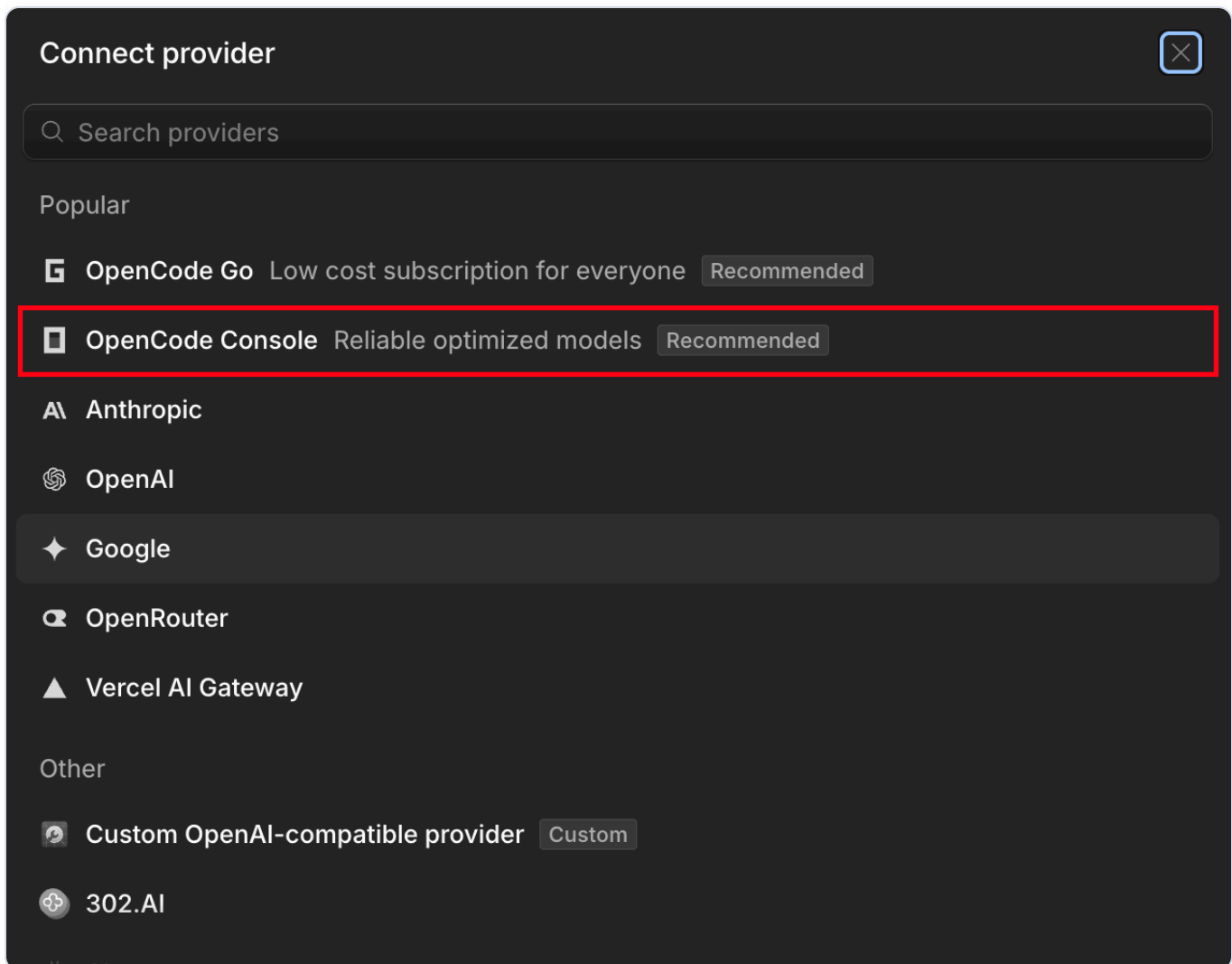
**Figure 2.** The invitation email. Click **Join workspace** and sign in with the same Tufts address the email was sent to.

**Use your Tufts email.** The invitation is tied to the address your instructor invited. If you sign in with a different email, you won't see the class workspace or be able to use the class budget.

## 4 Sign in to OpenCode Console

This step connects the app to your class account. You sign in — you do **not** need an API key or any password to copy.

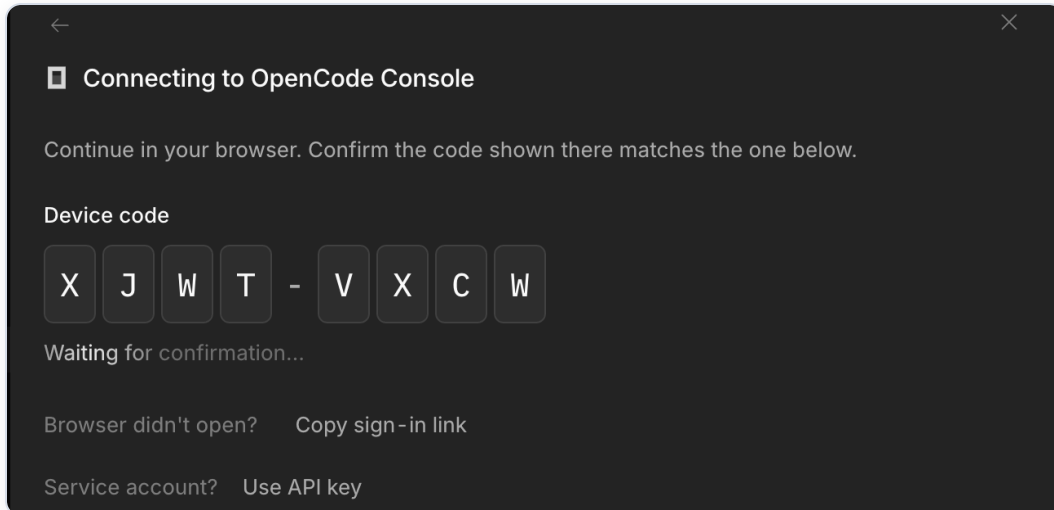
1. Open OpenCode. At the bottom is the message box — the same box you type questions in.
2. Type `/connect` and press `Enter`. A **Connect provider** list opens.
3. Click **OpenCode Console** — it's in the **Popular** section, marked **Recommended**.



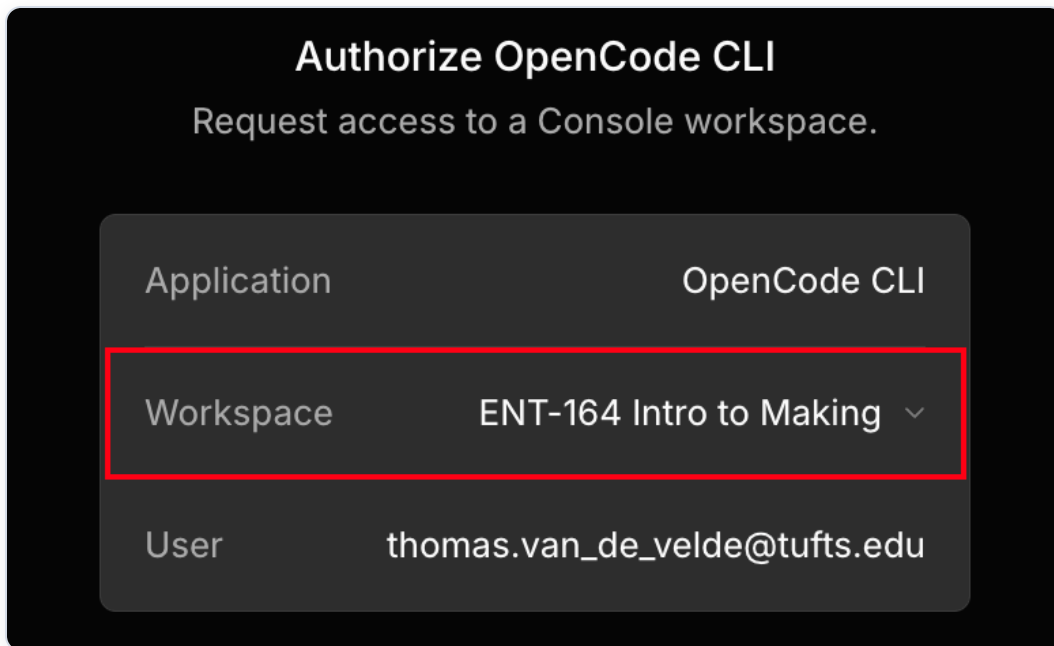
**Figure 3.** Typing `/connect` opens the provider list. Click **OpenCode Console** — outlined in red.

4. The app shows a **device code** and says "Continue in your browser". Your web browser opens the OpenCode sign-in page. If it doesn't open, click `Copy sign-in link` and paste it into your browser.
5. Sign in on that page with your **Tufts email**.

6. **Check the workspace.** The authorization page must show the class workspace, **ENT-164 Intro to Making**. If it shows a different workspace, click it and choose **ENT-164 Intro to Making** before you approve.
7. Confirm the code on the page matches the code in the app, and approve the sign-in.
8. Switch back to OpenCode. The window now says **OpenCode connected**, and OpenCode Console appears under **Connected providers**. You only do this once — the app remembers you.



**Figure 4.** The device code in the app. Your browser opens to the sign-in page; after you sign in there and approve, this dialog finishes on its own.



**Figure 5.** The authorization page in your browser. **Workspace** must say **ENT-164 Intro to Making** (outlined in red) — if it shows a different workspace, click it and choose the class one before you approve.

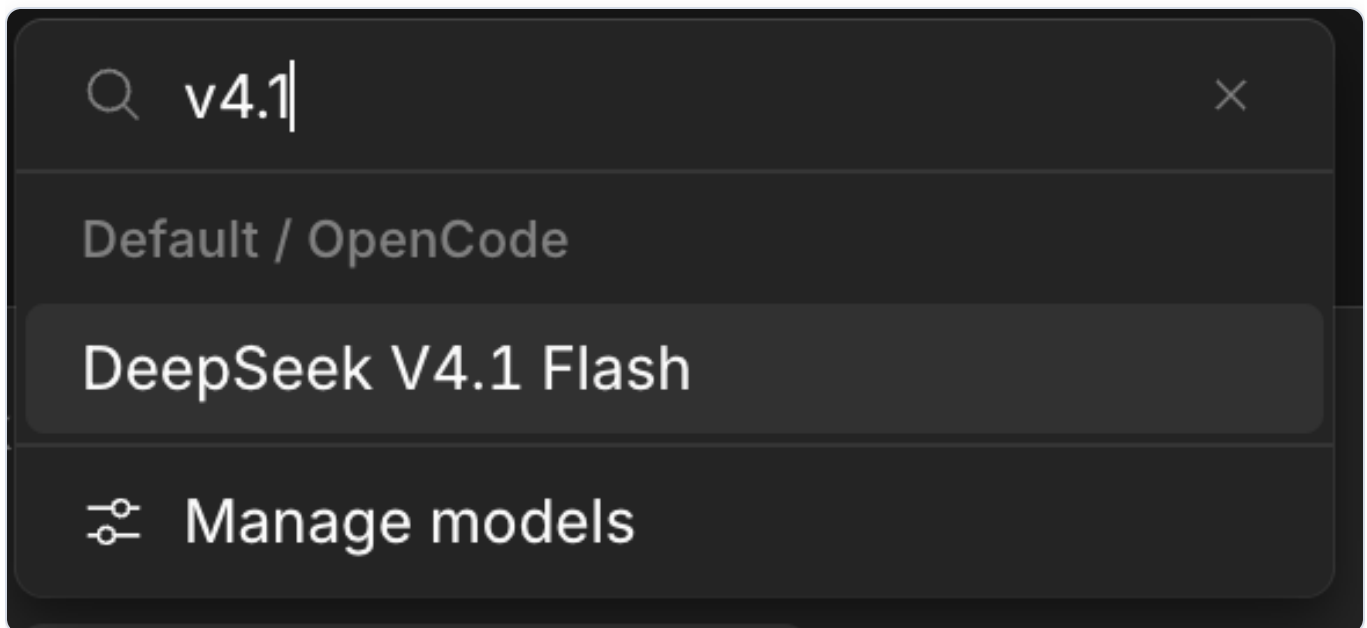
**Ignore "Use API key".** The dialog offers it under "Service account?" — that is for shared team setups, not for you. Students sign in with their account, so you never create or paste a key.

**Prefer menus?** The same provider list is at **Settings → Providers → OpenCode Console → Connect**. If your Settings sidebar shows a **Servers** list, open your server first — usually **Local Server** — and Providers is inside it.

## 5 Choose DeepSeek V4.1 Flash

Click the model button under the message box (it shows the model you're using, such as **Big Pickle** ), then:

1. Type **v4.1** in the search box at the top of the panel.
2. Click **DeepSeek V4.1 Flash** — it's listed under **Default / OpenCode**.
3. If it doesn't appear, click **Manage models** at the bottom, search **v4.1**, and turn on the switch next to **DeepSeek V4.1 Flash**. Then close that panel and pick it in the model list.

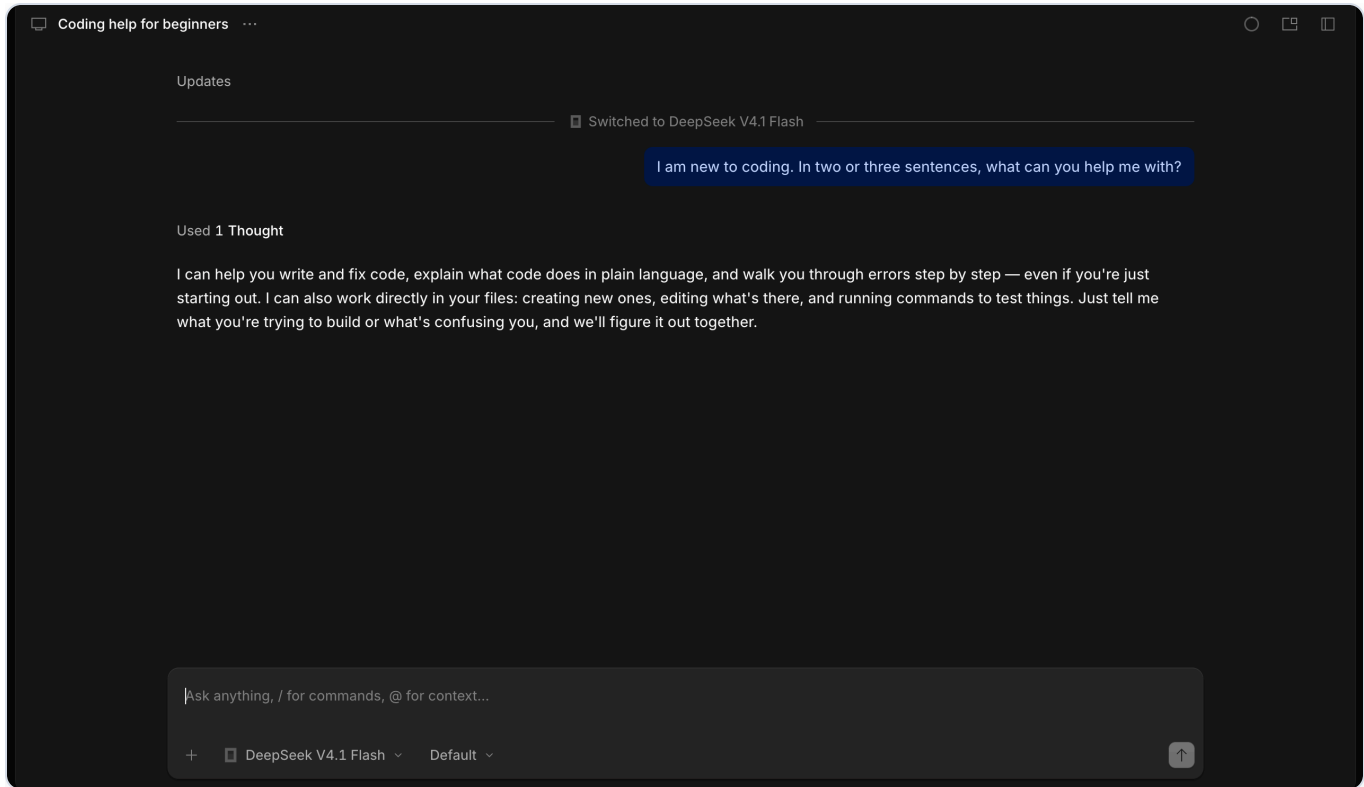


**Figure 6.** In the model list, type **v4.1** and click **DeepSeek V4.1 Flash**. **Manage models** is the last row if you need to switch it on first.

**Did it work?** The button under the message box now shows **DeepSeek V4.1 Flash**. That's the model every new question will use.

## 6 Ask your first question

1. Click the message box at the bottom that says "Ask anything...".
2. Type a question in plain English, for example: *"I'm new to coding. What can you help me with?"*
3. Press **Enter** (or click the arrow button at the right) and wait a few seconds.



**Figure 7.** Your question appears in the bubble on the right; the answer gets written below it. The model's name is always visible under the message box.

**To quit:** on a Mac, press **⌘** + **Q** ; on Windows, close the window or press **Alt** + **F4** . To come back later, just open OpenCode again — your conversations are saved as tabs at the top.

## 7 Add your class tools

OpenCode can switch on a set of **class tools** from the class website — extra abilities your instructor keeps up to date. The main one is a **wiring-diagram maker** for your ESP32 kit: describe a circuit and OpenCode draws a labelled breadboard picture with the jumper wires routed for you.

1. Click in the message box and paste this whole sentence:

**COPY AND PASTE THIS**

Add my class skills to opencode: <https://tuftmaker.github.io/ENT-164/skills/>

2. Press **Enter** and let it work. It downloads the class tools and notes where they came from, so it can update them later.
3. If it asks for permission — something like *"Can I change your settings file?"* — click **Allow** or **Yes** . It's only changing OpenCode's own settings, not anything else on your computer.
4. Quit OpenCode and open it again: on a Mac, press **⌘** + **Q** ; on Windows, close the window. The new tools only load at startup.

5. Test them. Type this into the message box and press **Enter** :

**TRY ASKING**

I am a maker. How can you help me?

If the answer mentions the class tools and the wiring diagrams, you're set. You can ask for a diagram any time — for example *"Draw me a wiring diagram for connecting an LED to my ESP32"* — and each one is saved in your project folder as a picture you can zoom and trace.

**One-time step.** The class tools update themselves when your instructor improves them — you never repeat this.

## If something goes wrong

| WHAT YOU SEE                                               | WHAT TO DO                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Your computer won't open the app                           | Mac: in Applications, right-click (Control-click) OpenCode and choose <b>Open</b> , or use System Settings → Privacy & Security → <b>Open Anyway</b> . Windows: in the installer warning, click <b>More info</b> → <b>Run anyway</b> .                                                                                                                     |
| No invitation email                                        | Check your Tufts inbox's spam folder, and confirm your instructor has your correct <code>tufts.edu</code> address. Invitations can be sent again.                                                                                                                                                                                                          |
| The invitation page says it belongs to a different account | Your browser is signed in as someone else. Click <b>Sign out and continue</b> on that page, then sign in with the Tufts address the invitation was sent to.                                                                                                                                                                                                |
| Can't find the provider panel                              | Type <b>/connect</b> in the message box — the provider list opens right there. (The same list is under <b>Settings → Providers</b> ; if Settings shows a <b>Servers</b> list, open your server first — usually <b>Local Server</b> — and Providers is inside it.)                                                                                          |
| OpenCode Console isn't in the list                         | It's in <b>Popular providers</b> near the top, marked <b>Recommended</b> . If you still don't see it, click <b>Show more providers</b> .                                                                                                                                                                                                                   |
| The browser doesn't open for sign-in                       | Click <b>Copy sign-in link</b> in the app, then paste it into your browser's address bar.                                                                                                                                                                                                                                                                  |
| "This sign-in request has expired"                         | That's just the device code timing out. Click <b>Try again</b> to get a fresh one, and finish the sign-in a little faster.                                                                                                                                                                                                                                 |
| Signed in but DeepSeek isn't in the model list             | You may have authorized a different workspace. Go back to <b>Settings → Providers</b> , <b>Disconnect</b> OpenCode Console, connect again, and this time switch the workspace to <b>ENT-164 Intro to Making</b> on the authorization page. If it still doesn't appear, use <b>Manage models</b> in the model panel, search <b>v4.1</b> , and switch it on. |

| WHAT YOU SEE                                      | WHAT TO DO                                                                                                                                                                    |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An error about credits or balance                 | Don't add credit yourself. Check that you signed in to the <b>ENT-164 Intro to Making</b> workspace, then tell your instructor — the shared class budget may be out of funds. |
| It asks for an API key                            | You don't need one. Close that dialog and use the account sign-in instead; the API key option is only for shared service accounts.                                            |
| OpenCode says it can't reach the website (Step 7) | Check you're connected to wifi — the class tools download from the class website the first time.                                                                              |
| The class tools don't seem to be on               | Ask OpenCode <i>"I am a maker. How can you help me?"</i> If it doesn't know the class tools, paste the sentence from Step 7 again, then quit and reopen OpenCode.             |
| No answer, or it seems stuck                      | Wait ten seconds — the first reply can be slow. Check your internet, then ask again. Quitting and reopening the app also helps.                                               |

## Words you'll see

|                             |                                                                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OpenCode</b>             | The app on your computer where you chat with the AI.                                                                                              |
| <b>OpenCode Console</b>     | The service that connects the app to a tested set of AI models. You reach it by signing in with your class account.                               |
| <b>Device code</b>          | The short code the app shows during sign-in. Your browser shows the same code, and you approve it there.                                          |
| <b>Workspace</b>            | The shared class area in OpenCode Console — here, <b>ENT-164 Intro to Making</b> . Choose it when the sign-in page asks.                          |
| <b>Class tools (skills)</b> | Extra abilities the class website teaches OpenCode — including the wiring-diagram maker. OpenCode downloads its own copy and keeps it up to date. |
| <b>Provider</b>             | The company or service that runs the AI connection — here, OpenCode Console.                                                                      |
| <b>Model</b>                | The specific AI "brain" you talk to — here, DeepSeek V4.1 Flash.                                                                                  |
| <b>Session / tab</b>        | One conversation. Each tab at the top of the app is a separate chat.                                                                              |

## Where everything lives

| WHAT YOU WANT                        | WHERE TO CLICK                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accept the class invitation          | The link in the email your instructor sent to your Tufts address                                                                                                                                                                                                                                                                                                                            |
| Sign in or manage your account       | <code>opencode.ai/console</code> — sign in with your Tufts email                                                                                                                                                                                                                                                                                                                            |
| Connect or fix your provider         | Type <b>/connect</b> in the message box → <b>OpenCode Console</b> (or <b>Settings → Providers</b> ; if Settings shows a <b>Servers</b> list, open your server first — usually <b>Local Server</b> )                                                                                                                                                                                         |
| Switch to the class workspace        | On the authorization page during sign-in, choose <b>ENT-164 Intro to Making</b>                                                                                                                                                                                                                                                                                                             |
| Start a new chat                     | The <b>+</b> next to the tabs at the very top                                                                                                                                                                                                                                                                                                                                               |
| Change the model                     | The model button under the message box (shows "DeepSeek V4.1 Flash")                                                                                                                                                                                                                                                                                                                        |
| Show all models                      | Model button → <b>Manage models</b>                                                                                                                                                                                                                                                                                                                                                         |
| Your class tools and wiring diagrams | The class tools live at <code>tuftsmaker.github.io/ENT-164/skills/</code> ; every diagram you make is saved in your project folder as a picture                                                                                                                                                                                                                                             |
| Quit the app                         | Mac:  +  · Windows: close the window or  +  |

## Things to try in your first week

You don't need to know any code to start. Ask in plain English:

| YOU CAN ASK                                                         | WHAT YOU GET                                                          |
|---------------------------------------------------------------------|-----------------------------------------------------------------------|
| <i>"I'm new to coding. What can you help me with?"</i>              | A plain-language list of what the AI can do, and where to start.      |
| <i>"Explain this like I've never seen it before"</i>                | Any file or concept broken down step by step, in beginner terms.      |
| <i>"Draw me a wiring diagram for connecting an LED to my ESP32"</i> | A labelled breadboard picture — once the class tools are on (Step 7). |
| <i>"Help me prepare my Onshape sketch for the laser cutter"</i>     | Checks and fixes so your DXF is ready for Nolog.                      |

Once you're comfortable, the official docs at [opencode.ai/docs](https://opencode.ai/docs) show what else OpenCode can do.